

Avni Bhardwaj

+91 9691385721 | avnibhardwaj01.ab@gmail.com | [LinkedIn](#) | github.com/avnibhardwaj1

EDUCATION

Narsee Monjee Institute of Management Studies

B.Tech, Computer Engineering | CGPA: 3.37 / 4.0

Indore, India

2022 – 2026

TECHNICAL SKILLS

Data Engineering & Cloud: AWS (S3, EC2, ECS, Aurora), Google Cloud, Docker, Kubernetes, CI/CD, ETL Pipeline Design, Data Modeling, Data Warehousing, Kafka/Redpanda, Apache Flink, ClickHouse, Redis, MinIO, Grafana, Relational (MySQL) & Vector/NoSQL (Qdrant) Databases

Programming Languages: Python, C++, SQL (MySQL), HTML/CSS, JavaScript

Backend & Software Engineering: FastAPI, REST APIs, Microservices, JSON, OOP, Git, Unix, Multithreading

AI / Machine Learning: Generative AI, Machine Learning, Deep Learning, NLP, LLMs, Agentic AI, RAG, LangChain, LlamaIndex, Prompt Engineering, Text Classification, Tokenization & Vectorization

Frameworks & Libraries: PyTorch, Scikit-Learn, NumPy, Pandas, Streamlit

Embedded Systems & IoT: ESP32, Embedded C, Sensor Interfacing, GPIO, I2C, Wi-Fi Connectivity, Blynk Cloud

EXPERIENCE

Flight Physics Capabilities Intern

Airbus

Jan 2026 – Jul 2026

Bangalore, India

- **Technologies:** Python, FastAPI, LlamaIndex, Qdrant, vLLM, Docker, Jenkins, GitHub Copilot, NetworkX, PySide6
- Architected FP-BOT, a Dockerized RAG pipeline deployed via Jenkins, using LlamaIndex, vLLM, and Qdrant to index documentation from 30+ tools for natural-language querying via FastAPI.
- Built AirSimuPy, a Python/PySide6 simulation platform leveraging NetworkX and NumPy to execute custom block-diagram data pipelines, reducing team reliance on MATLAB Simulink.

PROJECTS

Autonomous AI Research Agent | *Python, Pydantic AI, Gemini API, GitHub Actions, SMTP*

Aug 2026

- Architected a zero-maintenance, autonomous LLM agent using Pydantic AI and Gemini to orchestrate tools, evaluate content quality, and synthesize high-signal daily digests from 5+ feeds including Hacker News and arXiv.
- Engineered a serverless pipeline via GitHub Actions cron that intelligently filters noise from 100+ daily articles, saving an estimated 5+ hours of manual research per week with zero infrastructure costs.
- Built a production-grade, SOLID-compliant backend featuring dependency injection for secure `--dry-run` mock testing, 100% type-safe tool execution via Pydantic schemas, and strict iteration caps to prevent hallucination loops.

Personalized AI Financial Advisor | *Python, Unsloth, LangChain, Ollama, PyTorch*

Apr 2025

- Built a data pipeline to clean, transform, and load 70k+ financial records (Finance-Alpaca, FiQA) for fine-tuning Llama-3 (8B) via PyTorch-based Unsloth, and architected an Agentic RAG system (LangChain ReAct) that ingests real-time stock APIs (yfinance) and Kaggle CSV data into a Streamlit UI for automated financial recommendations.

Confidential Deep Learning Research (Under Review for publication) | *PyTorch, DenseNet201, SE-ResNet50* 2025

- Architected an ensemble deep learning framework integrating 4 advanced models (DenseNet201, SE-ResNet50, MobileNetV3-Large-100, and Xception) to optimize image classification and diagnostic accuracy.
- Engineered the end-to-end model pipeline in PyTorch, handling extensive data preprocessing, hyperparameter tuning, and rigorous model evaluation.

ACADEMIC & EXTRACURRICULAR ACHIEVEMENTS

- AWS Academy Graduate — Cloud Architecting (2025)
- Tableau Foundation Course — Intellipaat (2025)
- Finalist, IIM Indore Chaitanya — The Leadership Event (Top 10/1000+), 2024
- Finalist, Kavach Hackathon (Govt. of India Cybersecurity Hackathon), 2023